University of Florida, Department of Mathematics

Numerical Analysis PhD Exam, August 2025

Do 4 (four) of the first 5 problems (1–5) and 4 (four) of the last 5 problems (6–10).

1. Suppose A is Hermitian positive definite.

(a) Prove that each principal submatrix of A is Hermitian positive definite.

(b) Prove that an element of A with largest magnitude lies on the diagonal.

(c) Prove that A has a Cholesky decomposition.

2. Assume $A \in \mathbb{C}^{m\times m}$.

(a) Show that A has a Schur decomposition.

(b) If A has a collection of m linearly independent eigenvectors, show that A is diagonalizable.

3. Let $A \in \mathbb{C}^{m\times n}$, with $m \geq n$ and $\text{rank}(A) = p = n \geq 3$. Let $a_1, a_2, \ldots$ denote the columns of A.

(a) Using the modified Gram–Schmidt process, write out expressions for q_1, q_2, q_3, the first three columns of Q in the QR decomposition of A.

(b) Show the vector q_3 found in part (a) is orthogonal to both q_1 and q_2.

4. Let $A = U\Sigma V^*$ be the singular value decomposition of $A \in \mathbb{C}^{m\times n}$. Let u_j denote column j of U.

(a) Suppose $\text{rank}(A) = p < n < m$. Show $\{u_1, u_2, \ldots, u_p\}$ is a basis for $\text{Col}(A)$ and $\{u_{p+1}, u_{p+2}, \ldots, u_m\}$ is a basis for $\text{Null}(A^*)$.

(b) Suppose A is full rank and $x \neq 0$. Let σ_i, $i = 1, \ldots, n$, be the singular values of A. Show

$$\sigma_1 \geq \frac{\|Ax\|_2}{\|x\|_2} \geq \sigma_n > 0.$$

If you want to use the property that $\|A\|_2 = \sigma_1$, then you must prove that.

5. If $q_1, \ldots, q_n \in \mathbb{C}^m$ is an orthonormal basis for the subspace $V \subset \mathbb{C}^m$, prove that the orthogonal projector onto V is QQ^*, where Q is the matrix whose columns are the q_j.

6. Consider the fixed point iteration method $x_{k+1} = g(x_k)$, $k = 0, 1, \ldots$, for solving the nonlinear equation $f(x) = 0$. Consider choosing an iteration function of the form

$$g(x) = x - af(x) - b(f(x))^2 - c(f(x))^3,$$

where a, b, and c are parameters to be determined. Assume f is sufficiently differentiable and the corresponding iterations $x_{k+1} = g(x_k)$ converge to a unique fixed point z. Find expressions for the parameters a, b, and c in terms of functions of z such that the iteration method is of fourth order.

7. Consider the Lagrange polynomial approximation $p(x)$ of a function $f(x) \in C^2[a,b]$ at two points $x_0, x_1 \in [a,b]$. Determine $\alpha(x)$ explicitly in the following formula

$$f(x) - p(x) = \frac{f^{(n+1)}(\alpha(x)) \prod_{i=0}^{n}(x - x_i)}{(n+1)!}$$

for the case

$$f(x) = \frac{1}{x}, \qquad x_0 = 1, \qquad x_1 = 2.$$

8. Assume that $f \in C^3[a,b]$ and $x, x+h, x+2h \in [a,b]$. Determine constants c_1 and c_2 such that

$$\left| f'(x) - \frac{1}{h}\left[-\frac{3}{2}f(x) + c_1 f(x+h) + c_2 f(x+2h)\right]\right| \le ch^2 \max_{a \le x \le b} |f'''(x)|,$$

where $c > 0$ is a constant independent of h.

9. Let $Q_m(f)$ denote the m-point Gaussian quadrature rule over the interval $[a,b]$ and with continuous weight function $\rho(x) \ge 0$, that is,

$$Q_m(f) = \sum_{i=1}^{m} \alpha_i f(x_i) \approx J(f) = \int_a^b \rho(x) f(x)\, dx.$$

Show that, if a and b are finite and f is continuous, then $Q_m(f) \to J(f)$ as $m \to \infty$.

10. Consider numerically solving the initial value problem

$$y'(t) = f(t,y), \quad 0 < t \le t_f, \qquad \text{with } y(0) = \eta.$$

Assume f is sufficiently differentiable and let h denote the step size. Prove that the method

$$y_{n+3} + y_{n+2} - y_{n+1} - y_n = h(f_{n+3} + f_{n+2} + f_{n+1} + f_n)$$

is consistent but not convergent.